1.4 e Is Irrational, and So Is e^2

WORKSHEET

This section uses two questions. The first, from 1997, proves that e is irrational. The second, from 2024, uses a cleverer version of the same idea to prove that e^2 is irrational, which turns out to be genuinely harder.

Warm-up

STEP III 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q7

Main question

STEP III 2024, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Suppose $e = p/q$. Multiplying e by q gives an integer. What does multiplying by $q!$ do to each term of the series for e?

STAGE 1 The warm-up: e is irrational

Ask yourself Compare a_n and b_n term by term. Which is bigger?

Pause and try Now finish the proof. Suppose $e = p/q$. Which value of n should you choose?

STAGE 2 Why e^2 is harder

Ask yourself Try the same idea on e^2. Its series is $\sum 2^k/k!$. What goes wrong when you multiply by $n!$ and look at the tail?

STAGE 3 The two tails

Ask yourself For an alternating series $t_1 - t_2 + t_3 - \cdots$ with $t_1 > t_2 > t_3 > \cdots > 0$, why is the sum between 0 and t_1?

STAGE 4 Integers again

Ask yourself What is the sign of the first term of the tail, $k = 2n + 1$? How does the tail compare with $g(2n)$?

STAGE 5 The contradiction

Pause and try Now use the bounds from Stage 3. What happens to $qf(2n) + pg(2n)$ when n is very large?

What's really going on

How irrational is e?

Joseph Liouville proved in 1840 that e^2 is irrational, and in fact that it doesn't satisfy any quadratic equation with integer coefficients. In 1873 Charles Hermite went much further: he proved that e is *transcendental*, meaning it is not a root of *any* polynomial with integer coefficients. His proof is a sophisticated version of exactly the argument above: build an integer, show it is strictly between 0 and 1. Nine years later Ferdinand von Lindemann adapted Hermite's method to show that π is transcendental, finally proving that the ancient problem of "squaring the circle" with ruler and compass is impossible.

What we still don't know

For all the power of these methods, embarrassingly simple questions remain open. Is $e + \pi$ irrational? Is $e\pi$? Nobody knows, although at least one of them must be transcendental (exercise 4). Is $\zeta(3) = 1 + \frac{1}{8} + \frac{1}{27} + \cdots$ irrational? Yes: Roger Apéry astonished the mathematical world by proving it in 1978. Is $\zeta(5)$? Nobody knows.

Approximation

Behind these proofs lies a deeper idea: a number is irrational if it can be approximated by fractions *too well*. The partial sums of the series for e are fractions that get very close to e very quickly; a fraction can't be approximated by other fractions that well. This is the starting point of *Diophantine approximation*, a branch of number theory you can explore further in Section 2.9, where a different family of integrals produces even better fractions closing in on e.

Try this next

1 **Using $1/e$ alone.** Prove that e is irrational by supposing $1/e = p/q$ and using only the alternating series for $1/e$.

2 **$\sin 1$ is irrational.** Use $\sin 1 = 1 - \dfrac{1}{3!} + \dfrac{1}{5!} - \cdots$ to prove that $\sin 1$ is irrational. (Multiply by a suitable odd factorial.)

3 **Why not π?** Explain why this method doesn't work directly for π. Then look up Ivan Niven's one-page proof (1947) that π is irrational, which uses an integral instead of a series. It is within reach of a strong A-level student.

4 **One of them is transcendental.** HARDER Show that e and π are the roots of $x^2 - (e + \pi)x + e\pi = 0$. You may assume that a root of a polynomial whose coefficients are algebraic numbers is itself algebraic, and that π is not algebraic (Lindemann). Deduce that at least one of $e + \pi$ and $e\pi$ is transcendental.

Further reading

- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 8, "Some irrational numbers". Proves that e, e^2, e^4 and π^2 are irrational, using exactly these ideas.
- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). A short classic covering everything from $\sqrt{2}$ to transcendence.
- **Julian Havil, *The Irrationals*** (Princeton University Press). A readable history of irrational numbers.

The full guided solution is in Section 1.4:
https://beyondstep.keithmcnulty.org/ch01/e-irrational.html